SECOND SEMESTER B.Sc. DEGREE EXAMINATION, MAY 2018

(CUCBCSS–UG)

Complementary Course

PHY 2C 02—MECHANICS, RELATIVITY, WAVES AND OSCILLATIONS

Time : Three Hours

Maximum : 64 Marks

Section A

Answer **all** *questions.*
Each question carries 1 *mark.*
Answer in a word **or** *phrase.*

1. The centrifugal force acting on a particle of mass m, rotating with angular velocity $\vec{\omega}$ is :

   (a) $-m(\vec{\omega}\times\vec{r})$.

   (b) $-m\left(\vec{\omega}\times\frac{d\vec{r}}{dt}\right)$.

   (c) $-m\left(\frac{d\vec{\omega}}{dt}\times\vec{r}\right)$.

   (d) $-2m\left(\vec{\omega}\times\frac{d\vec{r}}{dt}\right)$.

2. When speed of rod along its length is increased, the length of rod :

   (a) Increases.

   (b) Decreases.

   (c) Remains unchanged.

   (d) Becomes zero.

3. The mass of an electron is double its rest mass then the velocity of the electron :

   (a) $\frac{c}{2}$.

   (b) $2c$.

   (c) $\frac{\sqrt{3}}{2}c$.

   (d) $\sqrt{\frac{3}{2}}c$.

4. Energy density for a plane harmonic wave is ———.

5. A train moving with constant velocity is :

   (a) An inertial frame.

   (b) A non-inertial frame.

   (c) Something inertial and sometimes non-inertial frame.

   (d) Neither inertial nor-inertial frame.

6. The total energy of a particle executing SHM is proportional to :
   (a) Displacement from equilibrium position.
   (b) Frequency of oscillation.
   (c) Velocity in equilibrium position.
   (d) Square of amplitude of motion.

7. The relativistic relation between momentum p and energy E is :
   (a) $E = \frac{p^2}{2m}$.
   (b) $E = p^2c^2 + m_0^2c^4$.
   (c) $E = \sqrt{p^2c^2 + m_0^2}c^4$.
   (d) $E = \frac{p^2}{2m} + m_0c^2$.

8. Which of the following is a Galilean invariant :
   (a) Velocity.
   (b) Acceleration.
   (c) Both of these.
   (d) None of these.

9. The motion of one projectile as seen from another projectile is :
   (a) A straight line.
   (b) A parabola.
   (c) A circle.
   (d) An ellipse.

10. At what speed the length of rod becomes half of its proper length :
   (a) $\frac{c}{2}$.
   (b) $\frac{c}{\sqrt{2}}$.
   (c) $\frac{\sqrt{3}}{2}c$.
   (d) $\sqrt{\frac{3}{2}}c$.

(10 × 1 =

Section B

Answer **all** *questions.*
Each question carries 2 *marks.*
Answer in **a short paragraph** — *three* **or** *four sentences.*

11. Name the types of frames of reference. Differentiate between them.
12. Give two examples of conservative and two examples of non-conservative forces.
13. What do you mean by time dilation ?
14. Explain the hypothesis of Galilean invariance.
15. What is centrifugal force ? Illustrate with example.
16. What is the significance of wave function ?
17. Why was the Michelson Moreley experiment performed ?

(7 × 2 =

Section C

Answer any **three** *questions.*
Each question carries 4 *marks.*
Answer in **a paragraph** *of about* **half a page to one page.**

18. Derive an equation for the energy density of a wave.
19. Write a note on electron microscope.
20. Show that motion of a particle under a central force takes place in a plane.
21. Explain the twin paradox.
22. Explain Lorentz Fitzgerald contraction and derive an expression for the same.

(3 × 4 = 12 marks)

Section D

Answer any **three** *questions.*
Each question carries 4 *marks.*
Problems-write all relevant formulas.
All important steps carry separate marks.

23. A particle of rest mass m is moving with a velocity 0.9c, calculate (i) its relativistic mass ; (ii) its kinetic energy.
24. The potential energy possessed by a particle moving under the influence of a conservative force is given by $U(x) = x^3 - 9x^2 + 24x$. Find the force on the particle.
25. A pendulum is of length 50 cm. Find its period when it is suspended in (i) a lift falling at a constant velocity of 5 m/s. (ii) a lift rising at a constant acceleration of 2 m/s.^{2}
26. Consider a ship moving with a uniform velocity of 18 m/s relative to the earth. Let a ball be rolled at a speed of 2 m/s. relative to the ship, in the direction of motion of the ship. Find the speed of the ball relative to the earth, according to Galilean transformations.
27. A young man goes to the pole star and comes back to the earth on a rocket. Calculate the age difference between him and his twin brother who preferred to stay on the earth. The rocket velocity $v = (4/5)\,c$ and the distance between the earth and the pole star is 40 light years. (Light year is a unit of distance, 1 light year = $3 \times 10^8 \times 60 \times 60 \times 24 \times 365$ m.)

(3 × 4 = 12 marks)

Section E

Answer any **two** *questions.*
Each question carries 8 *marks.*
(Essays. Answer in about **two** *pages).*

28. What is ether hypothesis ? Explain the Michelson Morley experiment.
29. Derive the time dependent Schrödinger equation.
30. Derive the differential equation of a particle executing simple harmonic motion. Also derive expression for its period, velocity and acceleration.
31. Mention the consequences of special theory of relativity and derive Einstein's mass energy relation.

(2 × 8 = 16 marks)